

REGULAR ARTICLE

Ecological patterns of variability in the life history of the European anchovy (*Engraulis encrasicolus*)

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Funding information

Ministry of Education and Science of the Russian Federation, Grant/Award Number: 124022400148-4

Abstract

The study of life-history variability is crucial in evolutionary theory and ecology, especially for commercial species impacted by fishery pressure and climate change. The European anchovy (*Engraulis encrasicolus*) is a key commercial fish in the East Atlantic and Mediterranean regions. This study aimed to identify life-history trends in this species and evaluate how environmental factors influence its life-history parameters. A meta-analysis of data from 23 areas within the anchovy's native range included parameters such as maximum observed total length, maturation length, length at age 1 year, maximum age, asymptotic length and growth performance index. Three environmental factors were examined: sea surface temperature, net primary production and sea surface salinity. Findings revealed that anchovies in lower latitudes and warmer environments had smaller maximum and maturation sizes and lower early growth rate. Temperature exhibited significant negative correlation with maximum and maturation size, but regions with low salinity, such as the Azov-Black Sea basin, deviated from this pattern. There was no significant association between temperature and lifespan across the entire study range. The correlation between productivity and life-history parameters was least pronounced. Hence, these findings suggest that ongoing climate warming may decrease anchovy size. However, temperature's influence on life history involves trade-offs with both benefits and costs for adaptability, suggesting its effects are not universally consistent.

KEYWORDS

age, climate change, environmental factors, growth, primary production, salinity, temperature-size rule

1 | INTRODUCTION

Life-history parameters such as size, lifespan, growth rates, timing and length of sexual maturity define the strategies that genes employ to maximize their replication under specific ecological conditions (Dawkins, 2016). Different environmental conditions will favour different traits and the alleles that determine them, leading to polymorphism. The study of the patterns of this variability is a fundamental problem in evolutionary theory and ecology. This problem becomes particularly relevant in the context of a rapidly changing climate, which poses a serious threat to Earth's marine ecosystems. In the 21st

century, the global average temperature is increasing at a rate seven times greater than that observed in the 20th century (Sheridan & Bickford, 2011). This rapid shift in temperature triggers three universal ecological responses: (1) the northward expansion of tropical species ranges, (2) alterations in phenology and behaviour, and (3) a reduction in the sizes of aquatic poikilothermic organisms (Gardner et al., 2011; Sheridan & Bickford, 2011). The latter response concerns overall life-history strategies and corresponds to the temperature-size rule (TSR), which postulates a decrease in maturation and maximum size while there is an increase in growth during the early stages of ontogeny (Atkinson, 1994). However, the mechanisms driving changes

in growth and size in response to warming remain poorly understood and are a subject of ongoing debate (Verberk et al., 2021). Investigating the ecological patterns of life-history variability could help address these knowledge gaps.

The European anchovy *Engraulis encrasicolus* (Linnaeus, 1758) is a pelagic, marine and brackish-water, short-lived species of fish belonging to the Engraulidae family. It inhabits depths of up to 400 m. Its range includes the Eastern Atlantic: from the shores of Norway in the north to the coast of South Africa in the south. It is found along all coasts of the Mediterranean, Black and Azov Seas, recorded in the Baltic Sea off the coast of Estonia (Froese & Pauly, 2024). Anchovies form large aggregations, reaching high abundance and biomass. They feed on planktonic organisms, serving as prey for a wide variety of fish species, birds and marine mammals; they play a key role in transferring energy from lower to higher trophic levels (Palomera et al., 2007). Within the Mediterranean basin, anchovy is a significant commercial species, ranking first in total catches at 342,000 tons in 2021 (FAO, 2023).

The study and monitoring of the biological characteristics of anchovies hold important applied and theoretical significance. A substantial amount of research has been dedicated to studying the life-history traits of Mediterranean anchovies. Data on the sex ratio, size-age structure, age and growth of anchovies in the western Mediterranean are presented in the works of Bellido et al. (2000), Basilone et al. (2004, 2017) and Bacha et al. (2010). Biological parameters of this species from the Adriatic Sea have been examined in the studies by Sinovčić (2000, 2004). Information on the age and growth of anchovies off the coasts of Tunisia (Khemiri et al., 2007), and waters of Egypt (El-Betar et al., 2023) and Greece (Machias et al., 2000) are available. Some aspects of the life history and the issue of commercial exploitation of anchovies from various regions of the Mediterranean have been analysed in the work of Colloca et al. (2013). Studies have also documented data on the growth of anchovies on the Atlantic coasts of Africa (Amponsah et al., 2016) and Europe (Błaszczuk, 1999; Ramos & Santos, 1999; Uriarte et al., 2016). Published literature includes data on biological indicators of this species within the Azov-Black Sea region (Chesalin et al., 2020; Samsun et al., 2004, 2006; Zuyev, 2019; Zuyev & Skuratovskaya, 2023). There is a data gap regarding the life history of anchovy along the west coast of Africa, with available data points in the coastal waters of Morocco (FAO, 2018) and the Gulf of Guinea off the coast of Ghana (Amponsah et al., 2016).

Anchovies exhibit a high degree of genetic polymorphism, which is closely linked to the diverse environmental conditions they inhabit (Bembo et al., 1996). For instance, within the Mediterranean Sea, two genetically and ecologically distinct anchovy ecotypes have been identified: the coastal ecotype (found near river mouths) and the marine ecotype (found in open waters) (Catanese et al., 2017). Although these ecotypes coexist in the same regions, they occupy different ecological niches. The authors propose that the divergence of these ecotypes may have arisen from allopatric speciation (geographic isolation) during the last glacial period, followed by secondary contact and partial reproductive isolation. The coastal ecotype is typically found

near river mouths, where conditions include low salinity and high productivity. In contrast, the marine ecotype exhibits greater ecological plasticity, enabling it to adapt to a variety of conditions, including oligotrophic waters (Catanese et al., 2017). As a result, salinity and feeding habits are considered key drivers of genetic divergence between these forms. However, it can be assumed that a broader range of environmental factors is involved in the formation of ecotypes, influencing both genotypic variability and phenotypic plasticity.

The extensive data on the biological parameters of anchovies from various parts of their range allow for attempts to establish the connection between the life-history parameters of this species and environmental factors. The study by Basilone et al. (2004) examines the ecological variability in the growth and production of anchovies from the Mediterranean Sea, the Bay of Biscay and the coast of South Africa in relation to sea surface temperatures (SSTs) and chlorophyll-a concentration. The authors note a negative relationship between maximum size and temperature and primary production (PP). In a more recent study (Basilone et al., 2017), anchovy growth from two different regions of the Mediterranean Sea is considered: the more productive Tyrrhenian Sea and the oligotrophic Sicilian Strait. The authors highlight a positive correlation between the sizes of yearlings and temperature, and a negative one with PP (Basilone et al., 2017). A later study (Basilone et al., 2020) focuses on the spawning ecology of the anchovy in the Strait of Sicily. The authors note that habitat differences before and during the spawning peak affected anchovy biology. Upwelling conditions associated with high productivity were a positive driver of anchovy growth and reproduction (Basilone et al., 2020). In the Mediterranean Sea basin, variability in the length at sexual maturity was studied. The obtained results showed a significant effect of the condition factor, indicating that fish in better condition reached maturity earlier (Ferrerri et al., 2021). A study on size variability among certain pelagic species in the Mediterranean Sea, including anchovy, found that the average sizes of these species are negatively correlated with temperature across the region (Hattab et al., 2021). The ecological and geographical variability of growth and sizes of eight species of the genus *Engraulis* has been studied in the work by Taboada et al. (2024). The authors note a decrease in the maximum sizes of this taxonomic group under high-temperature conditions at low latitudes, confirming the applicability of Bergmann's and James's rules (which postulate that within a broadly distributed taxonomic clade or within a species, body size tends to increase in colder climates). These promising studies should be continued and expanded to include other areas of the range and ecological factors. This knowledge will not only help uncover the mechanisms of variability in anchovy but will also enhance the theoretical foundation for the sustainable management of this economically important commercial species.

The Azov-Black Sea Basin is an especially interesting area for research. This region is connected to the Mediterranean Sea through the Bosphorus Strait; however, it significantly differs from it in hydro-meteorological characteristics: in the Black and Azov Seas, salinity is lower, surface water temperatures are lower and climate seasonality is higher. The Black and Azov Seas also differ significantly from each

other: the salinity of the Azov Sea is lower, while temperature fluctuations, productivity and the risk of hypoxia are higher (Berdnikov, Dashkevich, & Kulygin, 2019; Berdnikov, Kulygin, & Dashkevich, 2019). This has contributed to the formation of two ecological forms of anchovy within the Azov-Black Sea Basin that differ in size and growth rate, while being genetically homogeneous (Chashchin, 1996; Vodiasova & Abramson, 2017; Zuyev, 2019). Hence, the study of anchovies in the Azov-Black Sea Basin allows for the assessment of the effects of phenotypic plasticity.

If temperature is accepted as the primary factor influencing anchovy life history, then the TSR (Atkinson, 1994) and the broader ecological response of aquatic ectotherms to warming (Gardner et al., 2011; Sheridan & Bickford, 2011) predict a decrease in maturation length, maximum size, lifespan and growth performance from high-latitude (cold) to low-latitude (warm) waters. However, size of one-year-old fish is expected to be larger in warmer waters. Deviations from this overall pattern would suggest that either temperature influences anchovy differently than predicted by the TSR or non-temperature factors (biotic and abiotic) exert a strong, multifaceted influence on life-history strategies in specific parts of their range.

It was therefore hypothesized that if temperature is the principal driver, life-history parameters will align with TSR predictions along a latitudinal gradient (Figure S1a). Significant deviations will signify the influence of other factors. This study aims to identify range-wide trends in life-history variation and evaluate the impact of environmental drivers on these parameters.

2 | MATERIALS AND METHODS

2.1 | Study area and datasets

The work is based on the life history of anchovy (38 data points) from 23 areas of its native range. Data were sourced from peer-reviewed literature, with a focus on studies providing reliable and consistent methodologies. Sources classified as doubtful (e.g. those where asymptotic length L_{∞} deviated significantly by more than $\pm 1/3$ from the observed maximum length $L_{\max}$) were excluded. Older studies were cross-validated with modern datasets in terms of ageing and growth modelling techniques to ensure consistency. Differences in measurement methodologies across historical data and regions were addressed by converting standard length (SL) and fork length (FL) to total length (TL) using the following regression equations: $TL = 0.394 + 1.043FL$ and $TL = 0.242 + 1.141SL$ (Froese & Pauly, 2024). The study included anchovy populations from the northeast Atlantic, the Mediterranean Sea, the Azov-Black Sea basin, the coastal waters of Morocco and the Gulf of Guinea off the coast of Ghana (Figure 1).

Among the observed life-history parameters are the maximum observed total length ($L_{\max}$), length at age 1 year (TL_1), which characterizes early growth rate, L_{mat} , the total length at which 50% of individuals are mature, and $A_{\max}$, the maximum observed age, which measures lifespan. The analysis also included theoretical parameters, which are informative when empirical data are not available or when a

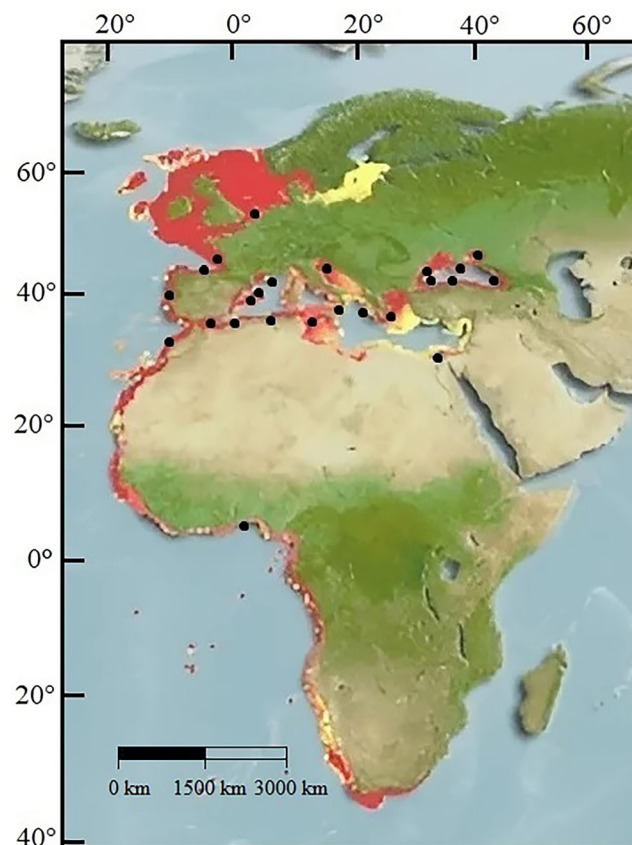


FIGURE 1 Map of the distribution of *Engraulis encrasicolus* (AquaMaps, 2019) and the regions included in this study (●).

population is highly exploited: L_{∞} , the asymptotic total length from the von Bertalanffy growth model; k coefficient, which describes how quickly an organism approaches its asymptotic length; and ϕ' , the growth performance index, which characterizes the growth rate. This index is calculated using the formula by Pauly et al. (1988): $\phi' = \lg k + 2 \lg L_{\infty}$. Most von Bertalanffy parameters were derived from otolith age readings, which are considered the gold standard for age determination in fish. However, four data points relied on length–frequency distribution analysis (LFDA) due to the lack of otolith data: the Gulf of Cádiz, waters off Morocco, Lake Manzala (Egypt) and the coast of Ghana (Table 1). The asymptotic lengths obtained from LFDA were validated by their close alignment with observed $L_{\max}$ values, ensuring their reliability for inclusion in the analysis.

Among the environmental factors influencing fish life-history parameters, temperature conditions (as a control factor), food availability (as a limiting factor) and salinity are highlighted (Brett, 1979). These environmental factors exhibit significant variation across the range of anchovy and have been selected for analysis. Temperature conditions are strongly correlated with latitude, so latitudinal trends in size and growth were examined as a first step in the analysis. SST and salinity datasets were sourced from the World Ocean Atlas (2025) (Table 2). Temperature data were aligned with sample collection periods. Net PP data were derived from the scientific literature (Table 3).

TABLE 1 Data on the biological parameters of *Engraulis encrasicolus* used in the analysis.

Region	Ageing and growth modelling method	Biological parameters	Source
North Sea, Dutch coast	Otoliths	L_{∞} , k , ϕ' , TL_1 , L_{mat} , L_{max} , A_{max}	Blaszczyk (1999)
Sea of Azov	Otoliths	L_{∞} , k , ϕ'	Pauly (1978)
Sea of Azov	Otoliths	L_{∞} , k , ϕ'	Volovik and Kozlitina (1983)
Sea of Azov	Otoliths	L_{∞} , k , ϕ' , TL_1 , L_{mat} , L_{max} , A_{max}	Chesalin et al. (2020)
Black Sea, northern part	Otoliths	L_{∞} , k , ϕ'	Pauly (1978)
Atlantic Ocean, Bay of Biscay	Otoliths	L_{∞} , k , ϕ' , TL_1	Guérault and Avrilla (1974)
Atlantic Ocean, Bay of Biscay	Otoliths	L_{∞} , k , ϕ' , TL_1 , L_{mat} , L_{max} , A_{max}	Uriarte et al. (2016)
Adriatic Sea, northern part	Otoliths	L_{∞} , k , ϕ'	ICES (2010)
Adriatic Sea, northern part	Otoliths	L_{∞} , k , ϕ' , TL_1 , L_{mat}	Sinovčić (2000)
Adriatic Sea, northern part	Otoliths	L_{∞} , k , ϕ' , A_{max}	STECF (2009a)
Atlantic Ocean, Portugal coast	Otoliths	L_{∞} , k , ϕ' , TL_1	Ramos and Santos (1999)
Atlantic Ocean, Spanish coast	Otoliths	L_{∞} , k , ϕ' , TL_1	Cendrero et al. (1981)
Black Sea, eastern part	Otoliths	L_{∞} , k , ϕ'	Pauly (1978)
Black Sea, western part	Otoliths	L_{∞} , k , ϕ'	Pauly (1978)
Black Sea, western part	Otoliths	L_{∞} , k , ϕ' , L_{max} , A_{max}	Yankova (2014)
Black Sea, southern part	Otoliths	L_{∞} , k , ϕ' , TL_1	Karacam and Düzgünes (1990)
Black Sea, southern part	Otoliths	L_{∞} , k , ϕ' , TL_1 , L_{mat} , L_{max} , A_{max}	Samsun et al. (2004)
Black Sea, southern part	Otoliths	L_{∞} , k , ϕ' , TL_1 , L_{mat} , L_{max} , A_{max}	Samsun et al. (2006)
Mediterranean Sea, Gulf of Lion	Otoliths	L_{∞} , k , ϕ' , TL_1 , A_{max}	Basilone et al. (2004)
Balearic Sea	Otoliths	L_{∞} , k , ϕ' , TL_1	Morales-Nin and Pertierra (1990)
Balearic Sea	Otoliths	L_{∞} , k , ϕ' , A_{max}	Colloca et al. (2013)
Aegean Sea	Otoliths	L_{∞} , k , ϕ' , A_{max}	STECF (2009b)
Ionian Sea, central part	Otoliths	L_{∞} , k , ϕ' , TL_1	Machias et al. (2000)
Alboran Sea	Otoliths	L_{∞} , k , ϕ' , A_{max}	Colloca et al. (2013)
Mediterranean Sea, Strait of Sicily	Otoliths	L_{∞} , k , ϕ' , TL_1 , L_{max} , A_{max}	Basilone et al. (2004)
Atlantic Ocean, Gulf of Cadiz	LFDA	L_{∞} , k , ϕ' , TL_1 , L_{max} , A_{max}	Bellido et al. (2000)
Mediterranean Sea, North Tunisian coast	Otoliths	L_{∞} , k , ϕ' , TL_1	Khemiri et al. (2007)
Mediterranean Sea, Algerian coast	Otoliths	L_{∞} , k , ϕ' , TL_1 , L_{mat}	Hémida (1987)
Mediterranean Sea, Bay of Benisaf	Otoliths	L_{∞} , k , ϕ' , TL_1 , L_{max} , A_{max}	Bacha et al. (2010)
Mediterranean Sea, South Tunisian coast	Otoliths	L_{∞} , k , ϕ' , TL_1	Khemiri et al. (2007)
Atlantic Ocean, waters of Morocco	LFDA	L_{∞} , k , ϕ' , L_{max}	FAO (2018)
Mediterranean Sea, Lake Manzala (lagoon)	LFDA	L_{∞} , k , ϕ' , TL_1 , L_{mat} , L_{max} , A_{max}	El-Betar et al. (2023)
Atlantic Ocean, Ghana coast	LFDA	L_{∞} , k , ϕ' , TL_1 , L_{mat} , L_{max}	Amponsah et al. (2016)

Abbreviations: L_{∞} , asymptotic length; k , growth constant in the Bertalanffy model; TL_1 , observed length at age 1 year; L_{max} , observed maximum TL; L_{mat} , TL at first maturity; A_{max} , observed maximum age; LFDA, length frequency data analysis.

2.2 | Data analysis

To analyse latitudinal trends in size and growth, regression analyses were performed. A Spearman rank correlation test was conducted to estimate the correlation between latitude and von Bertalanffy growth equation parameters alongside maximum length (L_{max}), length at maturity (L_{mat}), length at 1 year of age (TL_1) and maximum age (A_{max}). This non-parametric approach was selected for its suitability in handling potentially nonlinear patterns for ecological data (Quinn & Keough, 2002).

A robust multi-step statistical modelling framework was employed to assess the influence of SST, PP and salinity on key biological parameters of anchovy. The analysis was conducted in the R programming environment (R Core Team, 2025) using a suite of specialized packages.

The dataset consisted of up to 38 observations for the biological parameters (L_{∞} 38, k 38, ϕ' 38, TL_1 25, L_{mat} 14, L_{max} 17, A_{max} 19). Prior to model fitting, multicollinearity among the environmental predictors was assessed by calculating concavity (with values over 0.8 indicating

TABLE 2 Data on mean annual sea surface temperature (SST), and sea surface salinity used in the analysis (World Ocean Atlas, 2025).

Region, SST data period	SST, °C	Salinity, ‰
North Sea, Dutch coast, 1995–2004	11.5	35.0
Sea of Azov, 1955–1964	11.5	13.0
Sea of Azov, 1975–1984	13.5	13.0
Sea of Azov, 2015–2024	14.5	13.0
Black Sea, northern part, 1955–1964	14.5	18.0
Black Sea, eastern part, 1955–1964	16.5	18.0
Black Sea, western part, 1955–1964	14.5	16.0
Black Sea, western part, 1955–1964	15.5	16.0
Black Sea, southern part, 1985–1996	15.5	18.0
Black Sea, southern part, 1995–2004	15.5	18.0
Black Sea, southern part, 2005–2014	16.5	18.0
Atlantic Ocean, Bay of Biscay, 1965–1974	15.5	35.0
Atlantic Ocean, Bay of Biscay, 2015–2024	16.5	35.0
Adriatic Sea, northern part, 1995–2005	19.1	39.0
Atlantic Ocean, Spanish coast, 1975–1984	15.5	35.5
Atlantic Ocean, northern Portugal coast, 1995–2004	16.5	35.5
Mediterranean Sea, Gulf of Lion, 2005–2014	17.5	39.0
Balearic Sea, 1985–1994	18.5	39.0
Balearic Sea, 2005–2014	19.5	39.0
Aegean Sea, 2005–2014	18.5	39.0
Ionian Sea, 1995–2004	20.5	39.0
Alboran Sea, 2005–2014	19.5	39.0
Strait of Sicily, 1995–2004	19.5	39.0
Atlantic Ocean, Gulf of Cadiz, 1995–2004	19.5	36.5
Mediterranean Sea, Algerian coast, 1985–1994	19.5	39.0
Mediterranean Sea, Tunisian coast, 2005–2014	19.5	39.0
Atlantic Ocean, waters of Morocco, 2015–2024	20.5	37.5
Lake Manzala, Egypt, 2015–2024	23.5	4.0
Atlantic Ocean, Ghana coast, 2015–2024	27.5	35.0

high concavity) and by examining a correlation to identify pairs with $|r| > 0.7$.

To identify the most parsimonious and biologically plausible model for each response variable, a multi-model inference approach was used. The candidate set included generalized additive models (GAMs) to capture potential non-linear relationships. Given that the morphological data consisted of continuous, strictly positive values with a variance expected to increase proportionally with the mean, all models were specified using a gamma probability distribution with a log link function. This distributional assumption is biologically appropriate for size data as it inherently handles positive skew and heteroscedasticity, preventing physiologically impossible predictions

TABLE 3 Data on annual net primary production (PP) used in the analysis.

Region, PP data period	PP, gCm ⁻² year ⁻¹	Source
North Sea, Dutch coast, 1981–2004	170	Holt et al. (2012)
Sea of Azov, 2003–2011	247	Berdnikov, Kulygin, and Dashkevich (2019)
Black Sea, northern part, 1998–2007	144	Finenko et al. (2010)
Atlantic Ocean, Bay of Biscay, 1981–2004	151	Holt et al. (2012)
Adriatic Sea, 1997–2001	180	Bosc et al. (2004)
Atlantic Ocean, Spanish coast, 1981–2004	151	Holt et al. (2012)
Mediterranean Sea, Gulf of Lion, 1997–2001	189	Bosc et al. (2004)
Balearic Sea, 1997–2001	172	Bosc et al. (2004)
Aegean Sea, 1997–2001	125	Bosc et al. (2004)
Ionian Sea, 1997–2001	116	Bosc et al. (2004)
Alboran Sea, 1997–2001	230	Bosc et al. (2004)
Strait of Sicily, 1997–2001	153	Bosc et al. (2004)
Atlantic Ocean, Gulf of Cadiz, 1997–2001	230	Bosc et al. (2004)
Mediterranean Sea, Algerian coast, 1997–2001	149	Bosc et al. (2004)
Mediterranean Sea, Tunisian coast, 1997–2001	164	Bosc et al. (2004)
Atlantic Ocean, waters of Morocco, 1994–1995	285	Fischer et al. (2000)
Atlantic Ocean, Ghana coast, 1994–1995	200	Fischer et al. (2000)

(e.g. negative size) and providing a more robust fit than a normal (Gaussian) distribution would for such data (Zuur et al., 2009).

The following model structures were tested:

1. Additive models: Contained separate smooth terms for each predictor: $s(\text{SST}) + s(\text{PP}) + s(\text{salinity})$.
2. Interaction models: Built to test for non-linear synergistic effects using tensor product interactions between ecologically plausible predictor pairs: $s(\text{salinity}) + \text{te}(\text{SST}, \text{PP})$ and $s(\text{SST}) + \text{te}(\text{salinity}, \text{PP})$.

Given the limited sample size ($n \approx 38$), the maximum basis dimension for all smooth terms was restricted to prevent overfitting. The basis dimension (k) was set to a maximum of 3 for univariate smooths and 2 for tensor product interactions.

Model selection was performed using the corrected Akaike Information Criterion (AICc). All candidate models for a given parameter

TABLE 4 Growth parameters of *Engraulis encrasicolus* from different regions.

Region	Latitude, °N	L_{∞} , cm	k , year ⁻¹	t_0 , year	φ'	TL ₁ , cm	L_{mat} , cm	L_{max} , cm	A_{max} , year	Source
North Sea, Dutch coast	51	20.6	1.14	-0.10	2.22	14.9	15.0	22.5	4	Blaszczyk (1999)
Sea of Azov	46	11.7	1.16	-0.17	2.20	-	-	-	-	Pauly (1978)
Sea of Azov	46	13.3	0.61	-1.80	2.03	-	-	-	-	Volovik and Kozlitina (1983)
Sea of Azov	46	13.0	0.73	-0.86	2.09	9.7	9.7	14.4	3	Chesalin et al. (2020)
Black Sea, Northern Part	45	13.9	1.73	-	2.52	-	-	-	-	Pauly (1978)
Atlantic Ocean, Bay of Biscay	45	22.1	0.57	-	2.45	9.9	-	-	-	Guérault and Avrilla (1974)
Atlantic Ocean, Bay of Biscay	45	18.2	0.79	-1.00	2.42	13.2	13.7	17.9	4	Uriarte et al. (2016)
Adriatic Sea, northern part	44	15.9	0.51	-	2.11	-	-	-	-	ICES (2010)
Adriatic Sea, northern part	44	19.4	0.57	-	2.33	11.1	9	-	-	Sinovčić (2000)
Adriatic Sea, northern part	44	16.2	0.40	-2.04	2.02	-	-	-	5	STECF (2009a)
Atlantic Ocean, Portugal coast	43	15.8	0.53	-	2.12	11.1	-	-	-	Ramos and Santos (1999)
Atlantic Ocean, Spanish coast	43	21.3	0.48	-	2.34	10.4	-	-	-	Cendrero et al. (1981)
Black Sea, eastern part	43	14.1	1.01	-0.28	2.30	-	-	-	-	Pauly (1978)
Black Sea, eastern part	43	14.2	0.81	-0.54	2.21	-	-	-	-	Pauly (1978)
Black Sea, western part	43	15.0	1.26	-0.23	2.45	-	-	-	-	Pauly (1978)
Black Sea, western part	43	14.6	0.48	-1.54	2.01	-	-	14.7	4	Yankova (2014)
Mediterranean Sea, Gulf of Lion	42	19.1	0.35	-1.45	2.11	11.6	-	-	5	Basilone et al. (2004)
Black Sea, southern part	42	16.9	0.32	-1.99	1.96	10.3	-	16.9	3	Karacam and Düzgünes (1990)
Black Sea, southern part	42	17.1	0.28	-2.10	1.91	10.9	9.3	14.3	3	Samsun et al. (2004)
Black Sea, southern part	42	15.7	0.34	-2.53	1.92	10.4	9.3	14.3	3	Samsun et al. (2004)
Black Sea, southern part	42	18.9	0.16	-3.7	1.76	10.5	9.3	14.9	3	Samsun et al., 2006
Black Sea, southern part	42	16.8	0.23	-3.08	1.81	10.5	9.3	14.1	3	Samsun et al., 2006
Black Sea, southern part	42	18.5	0.22	-2.86	1.87	10.5	9.3	14.6	3	Samsun et al. (2006)
Black Sea, southern part	42	18.7	0.16	-3.97	1.75	10.5	9.3	14.9	3	Samsun et al. (2006)
Balearic Sea	41	19.1	0.35	-	2.11	11.0	-	-	-	Morales-Nin and Perterra (1990)
Balearic Sea	40	19.0	0.34	-2.32	2.09	-	-	-	5	Colloca et al. (2013)
Aegean Sea	38	19.1	0.38	-1.56	2.14	-	-	-	5	STECF (2009b)
Ionian Sea, central part	38	17.5	0.51	-	2.19	11.6	-	-	-	Machias et al. (2000)
Alboran Sea	37	19	0.34	-2.32	2.09	-	-	-	5	Colloca et al. (2013)
Mediterranean Sea, Strait of Sicily	37	18.6	0.29	-1.81	2.00	11.6	-	16.5	3	Basilone et al. (2004)
Atlantic Ocean, Gulf of Cadiz	37	19.0	0.90	-	2.51	-	-	18.5	4	Bellido et al. (2000)
Mediterranean Sea, North Tunisian coast ^a	36	20.4	0.32	-	2.12	11.6	-	-	-	Khemiri et al. (2007)
Mediterranean Sea, Algerian coast	36	16.6	0.59	-	2.21	11.9	11.0	-	-	Hémida (1987)
Mediterranean Sea, Bay of Benisaf	35	15.6	0.75	-1.32	2.26	12.9	-	17.0	4	Bacha et al. (2010)
Mediterranean Sea, South Tunisian coast ^a	34	18.3	0.36	-	2.08	9.8	-	-	-	Khemiri et al. (2007)
Atlantic Ocean, waters of Morocco	33	17.0	1.39	-0.15	2.60	-	-	17.0	-	FAO (2018)
Mediterranean Sea, Lake Manzala (lagoon)	31	12.5	0.95	-0.17	2.17	7.5	8.1	12.1	3	El-Betar et al. (2023)
Atlantic Ocean, Ghana coast ^b	5	12.8	0.58	-0.37	2.00	6.7	8.9	12.0	-	Amponsah et al. (2016)

Note: Data are ranked from high to low latitudes.

Abbreviations: L_{∞} , asymptotic length; k , growth constant in the Bertalanffy model; t_0 , age at which fish length in the Bertalanffy model is 0; φ' , growth performance index; TL₁, observed length at age 1 year; L_{max} , maximum TL; L_{mat} , TL at first maturity; A_{max} , maximum observed age.

^aFork length (FL) was the original measurement, values presented in the table are TL, converted using the formula $TL = 0.394 + 1.043FL$ (Froese & Pauly, 2024).

^bStandard length (SL) was the original measurement, values presented in the table are TL, converted using the formula $TL = 0.242 + 1.141SL$ (Froese & Pauly, 2024).

were ranked, and the full model with the lowest AICc score was selected as the best. For the best-performing model, the statistical significance of any interaction terms was investigated using a likelihood ratio test (LRT). This involved a comparison of the best model (e.g. with a $\text{te}(\text{SST}, \text{PP})$ interaction) against a nested additive model without the interaction.

The primary criterion for identifying an overfitted model was a deviance explained value exceedingly close to 1. Any model with a deviance explained value greater than 0.999 was automatically flagged as overfitted. This indicates the model accounts for almost all the variance in the data, including noise, which is a strong indicator of poor generalizability.

Following initial selection, the best full model underwent an iterative, stepwise backward selection to enhance parsimony. The non-significant terms ($p > 0.05$) were removed one at a time, but only if removal lowered the AICc value. This process continued until only significant terms ($p < 0.05$) remained or until further removal degraded the model fit.

The final models were validated through visual inspection of plots of observed versus predicted values, residuals versus fitted values, and Q-Q plots, and their overall performance was summarized using the deviance explained. Complete model validation and diagnostic procedures are provided in the Supporting Information. These diagnostics were used to confirm the adequacy of model fit and to ensure that the assumptions of the selected GAM structures were satisfied.

Analyses were conducted using the following R packages: *mgcv* (for GAMs; Wood, 2011), *tidyverse* (for data manipulation and visualization; Wickham et al., 2019), *MuMIn* (for model selection with AICc; Bartoń, 2025) and *visreg* (for visualization of partial effects; Breheny & Burchett, 2017).

3 | RESULTS

3.1 | Geographical patterns of anchovy life-history variability

Data on various life-history parameters (size, growth, maturation and age) ranging from high to low latitudes are presented in Table 4. The highest L_{max} value is noted in the North Sea ($\text{TL} = 22.5$ cm), while the lowest is found off the coast of Ghana ($\text{TL} = 12.0$ cm). The asymptotic length (L_{∞}) similarly varies from 20.6 cm in high latitudes to 12.5 cm in low latitudes. Additionally, TL_1 ranges from 14.6 cm off the coast of the Netherlands and 13.2 cm in the Bay of Biscay to 7.5 cm in Lake Manzala and 6.7 cm off the coast of Ghana. Length at maturity (L_{mat}) also tends to be higher at high latitudes, with a maximum value of 13.7 cm in the Bay of Biscay and a minimum value of 8.1 cm in Lake Manzala. There is a weak positive correlation between geographic latitude and size: for observed maximum length $r_s = 0.24$, for maturation length $r_s = 0.36$, for growth coefficient k $r_s = 0.26$ and for growth performance index $r_s = 0.20$. At the same time, TL_1 and the L_{∞} do not show a significant correlation with latitude ($r_s = -0.08$ and $r_s = 0.09$, respectively). The maximum age recorded is 5 years in

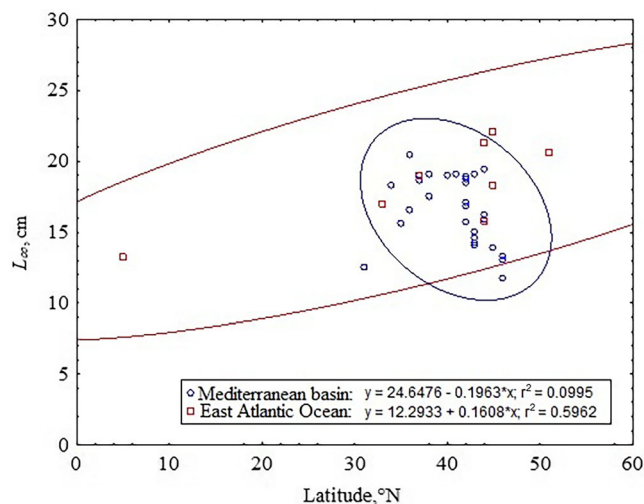


FIGURE 2 Asymptotic length of *Engraulis encrasicolus* related to latitude. Despite the pattern in the Atlantic Ocean showing an increase in length (red squares) with latitude, the Mediterranean basin (including the Black Sea and the Sea of Azov) exhibits a decrease (blue dots). The regression equations are provided in the box. Ellipses represent the 95% confidence areas.

some northern regions of the Mediterranean Sea, while in the North Sea and Bay of Biscay, it is 4 years. A lifespan of 3 years is typical in the Azov-Black Sea basin.

The general trend of increasing size from low to high latitudes does not follow a linear pattern. The asymptotic length of anchovies from the Azov and Black Seas is notably lower than expected based on the general pattern, resulting in an opposite local trend within the Mediterranean basin (Figure 2). Thus, in terms of maximum and maturation sizes, anchovies can be ranked as follows (from the largest to the smallest): Northeast Atlantic, Mediterranean Sea, Azov-Black Sea basin, Equatorial Africa.

3.2 | The role of temperature, production and salinity in the variability of anchovy life history

Latitudinal trends in life-history parameters are assumed to be associated with environmental factors and their heterogeneity throughout a species' range. Among the studied factors, the highest pairwise correlation was 0.39 (between SST and salinity), indicating low multicollinearity. Given this independence, 21 full (before parsimony analysis) candidate models were built to describe the potential relationships between the biological parameters and SST, PP, and salinity (Table S1). For each biological parameter, three GAMs using a gamma distribution were constructed. These included models with an additive structure as well as models with $\text{SST}:\text{PP}$ and $\text{salinity}:\text{PP}$ interactions.

Based on AICc values and AICc weights (which represent the relative probability that a model is the best among the candidate set) (Table S1), the best full models for most parameters (k , L_{max} , TL_1 , L_{mat} , A_{max}) were additive and followed the form:

TABLE 5 Approximate significance of smooth terms for the best full models.

Life-history parameter	Smooth term	EDF	RDF	F value	p value
L_{∞}	s(SST)	1.000	1.000	10.50	0.00272*
	te(S, PP)	3.239	3.449	18.19	3.4e-07*
k	s(SST)	1.232	1.411	0.335	0.759
	s(PP)	1.000	1.000	5.862	0.0210*
	s(S)	1.000	1.000	1.177	0.286
ϕ'	s(SST)	1.392	1.617	5.546	0.0144*
	te(S, PP)	5.822	6.587	5.104	0.000903*
$L_{\max}$	s(SST)	1.000	1.000	49.645	1.82e-05*
	s(PP)	1.000	1.000	0.235	0.637
	s(S)	1.838	1.974	22.411	6.71e-05*
TL_1	s(SST)	1.676	1.895	9.664	0.000852*
	s(PP)	1.000	1.000	0.000	0.999
	s(S)	1.000	1.000	18.189	0.000378*
L_{mat}	s(SST)	1.100	1.191	6.332	0.0228*
	s(PP)	1.000	1.000	0.829	0.384
	s(S)	1.000	1.000	9.356	0.0121*
$A_{\max}$	s(SST)	1.836	1.963	3.503	0.0787
	s(PP)	1.000	1.000	3.410	0.0877
	s(S)	1.851	1.967	14.58	0.000942*

Abbreviations: life-history parameters: L_{∞} , asymptotic length; k , growth constant in the Bertalanffy model; ϕ' , growth performance index; $L_{\max}$, maximum TL; TL_1 , observed length at age 1 year; L_{mat} , TL at first maturity; $A_{\max}$, maximum observed age; other terms: EDF, estimated degrees of freedom; RDF, reference degrees of freedom; SST, sea surface temperature; PP, primary production; S, salinity.

*Significant value ($p < 0.05$).

$$y \sim s(\text{SST}, k = 3) + s(\text{PP}, k = 3) + s(\text{salinity}, k = 3)$$

where $y \in \{k, L_{\max}, TL_1, L_{\text{mat}}, A_{\max}\}$. In contrast, L_{∞} and ϕ' (the latter being closely related to L_{∞}) were best modelled with salinity:PP interactions:

$$L_{\infty} \sim s(\text{SST}, k = 3) + \text{te}(\text{salinity}, \text{PP}, k = 2)$$

$$\phi' \sim s(\text{SST}, k = 3) + \text{te}(\text{salinity}, \text{PP}, k = 2)$$

The interactions were statistically significant based on an LRT comparing the interaction models to the best additive models ($p = 0.004$ for L_{∞} and $p = 0.003$ for ϕ').

Despite having the lowest AICc values, the interaction models for L_{∞} and ϕ' exhibited high concurvity (0.992) between the spline term s(SST) and the tensor product term te(salinity, PP), indicating severe multicollinearity. This makes it difficult to determine whether SST or the salinity:PP interaction is the primary driver of the changes in L_{∞} and ϕ' in the interaction model. However, the additive model for $L_{\max}$ (an empirical analogue of L_{∞}) can be referenced, which indicates a clear relationship with SST (Table 5), suggesting that SST is likely the dominant factor influencing maximum length.

The significance of smooth terms in the best-performing models (Table 5) indicates that SST and salinity have a greater influence on $L_{\max}$, L_{mat} and TL_1 rate than PP. PP was only a significant predictor for

k and, as part of a tensor product with salinity, for L_{∞} and the ϕ' . Conversely, SST was a non-significant predictor for both k and $A_{\max}$.

The parsimony analysis indicated that removing the non-significant PP predictor ($p > 0.05$) lowered AICc values for the models of TL_1 , L_{mat} and $L_{\max}$. Conversely, removing the SST predictor was supported for the models of k and $A_{\max}$ (Table 6). The explained deviance of the best models varied from 0.22 for k to 0.89 for $L_{\max}$.

The selected GAMs showed differing performance in describing the data for each life-history trait (Table 6 and Figures S2–S4). The GAM for k had the lowest deviance explained (0.22), indicating a high dispersion of k values even within the same region and under similar environmental conditions, which made it difficult to model. For example, in the Black Sea, k varied from 0.16 to 1.73. The best-performing models were for $L_{\max}$ and L_{mat} (with deviance explained values of 0.890 and 0.758, respectively). The other parameters demonstrated moderate values, ranging from 0.570 for ϕ' to 0.694 for $A_{\max}$.

According to the GAMs for von Bertalanffy growth parameters (L_{∞} , k , and ϕ'), SST showed strong negative effects on both L_{∞} and ϕ' (Figure 3a,e). These relationships did not significantly deviate from linearity (estimated degrees of freedom [EDF] ≈ 1 for L_{∞} , EDF = 1.39 for ϕ'). PP had a positive effect on k , while the effect of salinity was very weak (Figure 3c,d). A significant interactive effect was found between salinity and PP on L_{∞} and ϕ' . Specifically, lower salinity combined with high PP caused a reduction in L_{∞} and ϕ' , showing a

TABLE 6 Best-selected generalized additive models (GAMs) describing the influence of sea surface temperature (SST), primary production (PP) and salinity (S) on the life-history parameters of *Engraulis encrasicolus*.

Parameter	Formula	AICc	BIC	Deviance explained	Maximum concavity
L_{∞}	$\sim s(\text{SST}, k = 3) + \text{te}(\text{S}, \text{PP}, k = 2)$	157.3	164.7	0.646	0.992
k	$\sim s(\text{PP}, k = 3) + s(\text{S}, k = 3)$	16.5	21.8	0.220	0.130
ϕ'	$\sim s(\text{SST}, k = 3) + \text{te}(\text{S}, \text{PP}, k = 2)$	-9.10	-1.00	0.570	0.992
$L_{\max}$	$\sim s(\text{SST}, k = 3) + s(\text{S}, k = 3)$	55.8	54.5	0.890	0.269
TL_1	$\sim s(\text{SST}, k = 3) + s(\text{S}, k = 3)$	87.9	90.9	0.657	0.235
L_{mat}	$\sim s(\text{SST}, k = 3) + s(\text{S}, k = 3)$	57.6	50.8	0.758	0.186
$A_{\max}$	$\sim s(\text{PP}, k = 3) + s(\text{S}, k = 3)$	34.8	35.6	0.694	0.307

Note: Model fit was compared using the Akaike Information Criterion corrected for small sample sizes (AICc) and the Bayesian Information Criterion (BIC). Abbreviations: life-history parameters: L_{∞} , asymptotic length; k , growth constant in the Bertalanffy model; ϕ' , growth performance index; $L_{\max}$, maximum TL; TL_1 , observed length at age 1 year; L_{mat} , TL at first maturity; $A_{\max}$, maximum observed age.

significant non-additive interaction (Figure 3b,f). When salinity reached 25‰–30‰, the influence of PP on L_{∞} and ϕ' diminished. Simultaneously, low PP values slightly mitigated the positive effect of salinity on L_{∞} . The interaction plot for ϕ' shows a negative effect at both high and low PP levels when salinity is low. This pattern reflects the positive relationship of PP with k and its negative relationship with L_{∞} .

Similar to L_{∞} , its empirical analogue $L_{\max}$ exhibits a strong, effectively linear negative relationship with SST (EDF ≈ 1) and a non-linear positive relationship with salinity (EDF = 1.88). The effect of salinity was negligible below 20‰ and positive above that value (Figure 4a,b). However, $L_{\max}$ differs from L_{∞} : its variability is best described by a two-factor additive model without an salinity:PP interaction. TL_1 (reflecting early growth rate) also shows a negative overall trend with a non-linear response to temperature (EDF = 1.73). Between 14 and 21°C (the Mediterranean Sea basin), temperature had a weak effect with strong negative correlation after 21°C. In contrast, the relationship with salinity was linear (EDF ≈ 1) and positive (Figure 4c,d). The patterns for L_{mat} resemble those of $L_{\max}$ and TL_1 in their general trends, with a negative effect from SST (especially below 20°C) and a positive effect from salinity above 15‰–20‰. However, both relationships were non-linear (EDF = 1.64 and 1.57, respectively) (Figure 4e,f). PP had no significant effect on size or growth empirical parameters in the best selected GAMs but showed a weak positive effect on $A_{\max}$, as did salinity (Figure 4g,h). SST was not a significant predictor for $A_{\max}$ and was excluded during model selection based on parsimony.

The analysis of observed environmental influences on life-history traits led to the conclusion that colder waters were associated with higher values for maximum size, length at first maturity and growth performance index. This aligns with expectations based on the TSR and the hypothetical scheme for anchovy growth trajectory variability. In contrast, the early growth rate (TL_1)—which the TSR predicts should correlate positively with SST—exhibited a negative relationship in this study (Figure S1). It was anticipated that lifespan ($A_{\max}$) would demonstrate a negative association with

temperature; however, the observed pattern excluded SST as a significant predictor.

4 | DISCUSSION

4.1 | Temperature as a key driver of anchovy growth and size variability

The variability in the life history of anchovies should be considered complex and multifactorial; however, temperature is likely a key factor influencing the geographical variability of this species. This conclusion is supported by the positive correlation of maximum size and maturation length with latitude and, more strongly, by the GAMs (Figures 3 and 4), which showed a negative influence of temperature. In their study, Basilone et al. (2004) assessed the influence of temperature and production (chlorophyll-a concentrations) on anchovy growth for 14 regions across the Mediterranean Sea and the Atlantic Ocean. Within the studied range, a negative relationship between maximum sizes and temperature was observed, which is consistent with the findings of this study. Similar results were also reported by Hattab et al. (2021) in their study on the correlation between average sizes of five pelagic species in the Mediterranean Sea, including anchovy. The correlation was negative, with a trend of size reduction at a rate of 3.01% per 1°C.

The models indicated a negative effect of temperature on early growth rate across the entire study range: the largest yearlings were found in cold Dutch waters (Błaszczuk, 1999), whereas the smallest occurred in warm equatorial Africa (Amponsah et al., 2016) and Lake Manzala (El-Betar et al., 2023). Although a positive correlation between TL_1 and temperature in the Mediterranean Sea was reported by Bacha et al. (2010), no significant positive relationship between early growth rate and temperature was found across the broader Mediterranean region, including the Black and Azov Seas (Figure 4). It appears that early growth may depend on other complex factors, such as food web structure or the likelihood of extreme events like

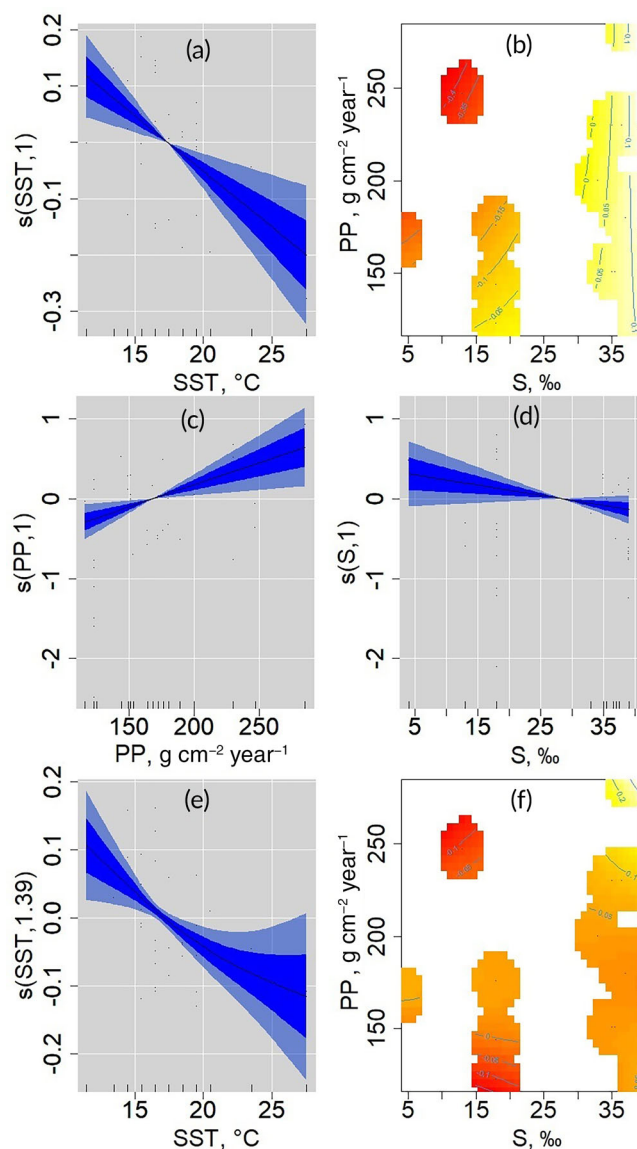


FIGURE 3 Generalized additive models (GAMs) of *Engraulis encrasicolus* von Bertalanffy parameters (L_{∞} , k) and growth performance index (ϕ') against sea surface temperature (SST), primary production (PP) and salinity (S). The models are as follows: (a, b) L_{∞} with a tensor product interaction $te(S, PP)$; (c, d) k with an additive model; (e, f) ϕ' with a tensor product interaction $te(S, PP)$. Values in brackets indicate the estimated degrees of freedom (EDFs). Blue shaded areas represent the 95% confidence bands. $g\ cm^{-2}\ year^{-1}$.

heatwaves. For example, the growth rate during the first year of life may depend on size-dependent predation pressure and the availability of appropriately sized food (Peters, 1986). Heatwaves and the associated hypoxia in regions with relatively low SST (such as Azov Sea) may lead to the selection of smaller individuals across all age groups (Cheung et al., 2013), including yearlings.

Conversely, no significant correlation was found between temperature and maximum observed age. This may be explained by the high commercial value of anchovies and varying fishing mortality rates across regions, which could truncate the age structures before

temperature-related effects on lifespan become apparent (Anderson et al., 2008).

4.2 | Mechanisms underlying temperature–size relationships

Experimental and field observations indicate that environmental temperature has a significant influence on the growth characteristics, development and sizes of aquatic ectothermic animals (Brett, 1979; Atkinson, 1994; Verberk et al., 2021). Apparently, the typical response to decreasing temperature from low to high latitudes is a slowdown in development and growth rate with an increase in maximum sizes (Bergmann's rule, James's rule, TSR) (Atkinson, 1994; Audzijonyte et al., 2020; Pauly, 1998; Verberk et al., 2021). Currently, the mechanisms underlying these patterns are a topic of significant debate. The first set of explanations focuses on the relationship between metabolic rate and temperature, alongside physiological constraints linked to the ratio of absorptive surface areas to body volume. This is exemplified by the gill oxygen limitation theory (GOLT) (Pauly, 2021; Pauly & Cheung, 2018). This theoretical framework applies to both phenotypic plasticity and genetic variability, termed the 'ghost of oxygen-limitation past' (Verberk et al., 2021). According to GOLT, elevated temperatures result in increased catabolism and oxygen requirements, which pose greater challenges for larger organisms due to their lower gill surface area-to-body volume ratio. Consequently, smaller individuals, which have shorter life cycles and maturation periods, may gain an adaptive advantage. The second set of explanations pertains to mechanisms concerning the allocation of resources between growth and reproduction (Kozłowski et al., 2004). Within this framework, growth and maximum size are constrained by reproductive expenditures, which tend to be higher in warmer environments due to increased mortality rates.

However, as these concepts operate at different levels—the GOLT at the physiological level, and the 'ghost of oxygen-limitation past' and energy allocation theory at the evolutionary level—they do not contradict each other. Instead, different mechanisms likely contribute to temperature-induced selection or phenotypic responses, which highlights the complexity of temperature–size relationships.

4.3 | The Azov-Black Sea anchovy: an exception to the rule and the role of salinity

The variability in anchovy sizes generally aligns with the understanding of how temperature affects aquatic ectotherms. However, there is a notable exception, the anchovy of the Azov-Black Sea basin. Its size and growth rate are significantly lower than expected for the annual SST of the basin (Table 4). The Black Sea is an intracontinental semi-enclosed water body connected to the more saline Mediterranean Sea by the single narrow Bosphorus Strait and to the less saline Sea of Azov by the Kerch Strait. Its intracontinental location, combined with large volumes of river discharge and specific precipitation patterns,

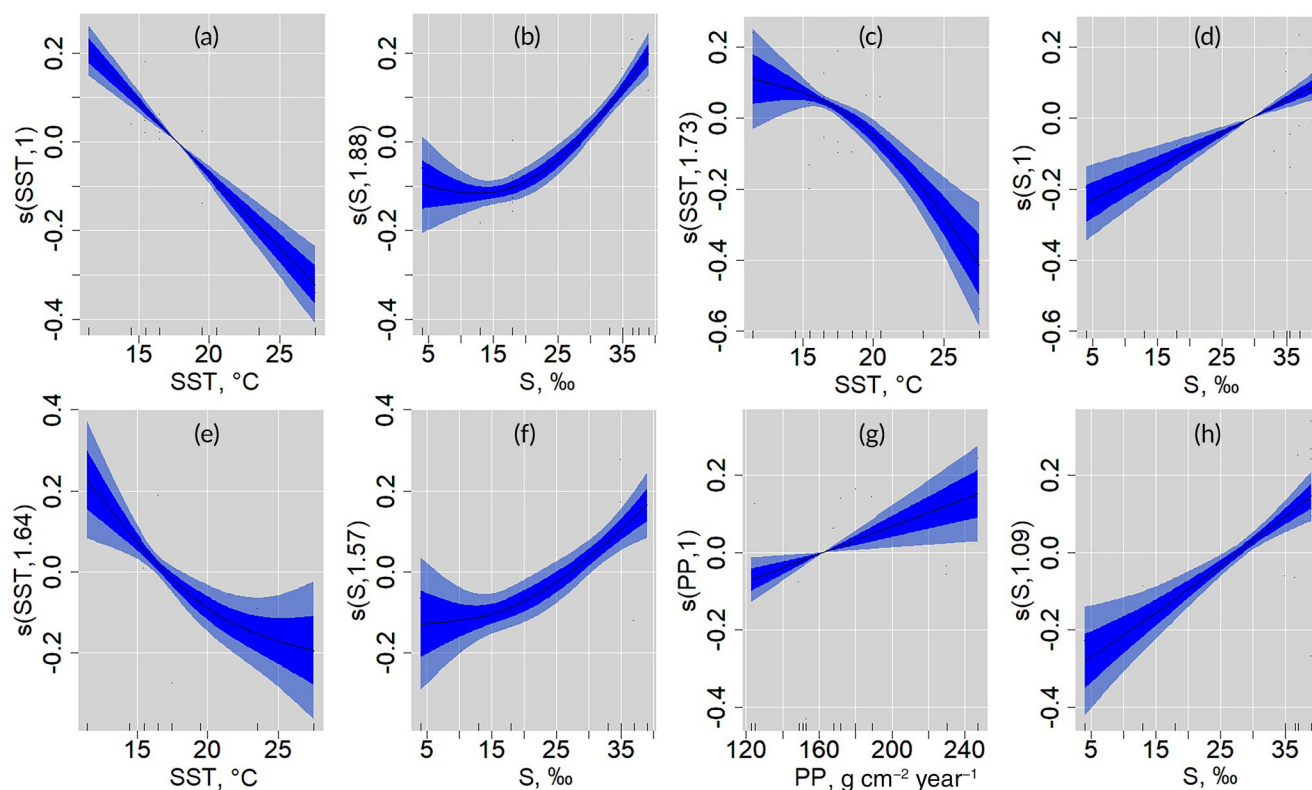


FIGURE 4 Generalized additive models (GAMs) of *Engraulis encrasicolus* life-history parameters ($L_{\max}$, TL_1 , L_{mat} , $A_{\max}$) against sea surface temperature (SST), primary production (PP) and salinity (S). The models are as follows: (a, b) $L_{\max}$; (c, d) TL_1 ; (e, f) L_{mat} ; (g, h) $A_{\max}$. Values in brackets indicate the estimated degrees of freedom (EDF). Blue shaded areas represent the 95% confidence bands.

contributes to its low salinity levels (17‰–18‰) and cold climate, which exhibits pronounced seasonality compared to the Mediterranean Sea (Berdnikov, Dashkevich, & Kulygin, 2019; Berdnikov, Kulygin, & Dashkevich, 2019). Seasonality, particularly when associated with low SST, can significantly influence growth, as selective pressure during winter is especially strong on smaller individuals, which deposit fewer lipids, which are critical for energy storage and survival under harsh conditions (Schultz & Conover, 1997). On the other hand, the cold winter months stimulate energy-consuming migrations of anchovy in the Azov-Black Sea basin (Chesalin et al., 2020; Gücü et al., 2017), which can lead to reduced growth rates and sizes. Additionally, the Sea of Azov is characterized by heatwaves in the summer months and periods of oxygen deficiency (Berdnikov, Dashkevich, & Kulygin, 2019; Berdnikov, Kulygin, & Dashkevich, 2019), which favours smaller individuals over larger ones due to greater tolerance to hypoxia (Verberk et al., 2021). It is noteworthy that within the Azov-Black Sea basin there are two genetically homogeneous but morphologically distinct forms (Vodiasova & Abramson, 2017; Zuyev, 2019). The Azov anchovy is characterized by smaller sizes (Chesalin et al., 2020), demonstrating pronounced phenotypic plasticity caused by the hydrometeorological features of the Sea of Azov (heatwaves and episodes of hypoxia). Furthermore, the models revealed a significant positive influence of salinity on anchovy size and growth. The example of the anchovy from Lake Manzala shows that low and unstable salinity, a characteristic shared by the

Azov-Black Sea basin, is associated with smaller sizes than those in marine environments.

Salinity influences fish growth primarily through osmoregulation, metabolic costs and species-specific tolerances. Optimal salinity levels promote efficient growth, while unsuitable salinity can lead to stress, reduced feeding and stunted growth. Low salinity values result in higher energy expenditure for osmoregulation, which may contribute to smaller body sizes (Boëuf & Payan, 2001). On the other hand, the anchovy species exhibits a wide salinity tolerance range, thriving in environments from brackish to marine waters. The coastal ecotype of anchovy is typically associated with low-salinity zones in estuaries (Catanese et al., 2017). It can be hypothesized that smaller fish, with their higher surface-area-to-volume ratio, achieve more efficient osmoregulation in fluctuating salinity environments like brackish water. As a result, they may adapt more quickly to salinity changes compared to larger fish. From this point of view, small body sizes in brackish water should be considered not as a stress response, but as an adaptive mechanism reinforced by natural selection. To confirm this hypothesis, it is advisable to conduct experimental studies involving the rearing of marine and coastal ecotypes of anchovies under varying salinity conditions. Such studies may also uncover additional physiological and molecular mechanisms underlying the development of euryhalinity in this species.

Overall, the example of the Azov-Black Sea anchovy shows that the life-history strategy is determined by a multitude of factors

affecting the intensity of phenotypic response and selection pressure. On one hand, comparatively low temperatures should promote selection aimed at increasing sizes, while on the other hand, a number of other factors (high lability of habitat conditions, seasonality, heat-waves and hypoxia, and possibly low salinity) provide more advantages to smaller individuals, or more precisely to the gene alleles determining smaller sizes. In other words, the costs of increasing in size under cold conditions may outweigh the potential benefits.

4.4 | Temporal variability

As an additional evidence indicating the relationship between anchovy size and temperature, the dynamics of its length over time can be considered, particularly in the context of global warming. Mayorova (1951) provides data on the sizes of anchovies in the northern part of the Black Sea in the mid-20th century when fishing was not yet developed and SST was significantly lower (Sakalli & Baştusta, 2018). According to her published data, the maximum standard length of anchovies in 1948–1949 reached 14.0 cm (or TL = 16.7 cm). By the end of the second year of life (1+), the total length of individuals ranged from 10.2 to 14.9 cm, with a weight of up to 20 g (Mayorova, 1951), significantly higher compared to current values. Yet, the age structure was similar to the present: the population was mainly composed of age groups 1+ and 2+, with the 3+ group rarely exceeding 1% (Mayorova, 1951). In the 21st century, from 2010 to the present, the maximum total length of anchovies varied from 13.2 to 14.7 cm (Zuyev, 2019; Zuyev & Skuratovskaya, 2023). Thus, over the past 75 years, a consistent reduction in the sizes and growth rate of Black Sea anchovies in the northern part is occurring. Taboada et al. (2024) come to similar conclusions regarding the temporal variability of anchovy growth in the Bay of Biscay. In their study, they note a decrease in the length and mass of anchovies across all age groups over a 30-year period (1990–2021), linking this effect to climate warming. However, it should be noted that prolonged and high fishing pressure can also lead to a decrease in size (Altukhov, 2001; Ricker, 2010), thereby masking the potential impact of rising temperatures.

4.5 | Productivity and growth: a contentious relationship

Based on the results, the influence of productivity on the growth and size of anchovies is less pronounced than that of the other factors considered. Specifically, productivity was removed from the full models for $L_{\max}$, TL_1 and L_{mat} due to its low significance and for the sake of model parsimony (Tables 5 and 6). Only a weak effect of PP on k was found—especially considering the low deviance explained by its model (0.22)—and a practically non-significant effect on $A_{\max}$ was observed (Figure 4). This might suggest that anchovies reach their asymptotic length faster in more productive ecosystems and may live slightly longer. However, interpreting k as a parameter characterizing

growth rate is challenging because, for different values of L_{∞} , the same value of k can correspond to different absolute growth rates. Therefore, concluding that anchovies have a higher growth rate in more productive ecosystems without analysing other parameters would be unreliable.

At the same time, productivity influenced L_{∞} and ϕ' as part of a tensor product, reducing their values when productivity was high, especially under conditions of low salinity (Figure 3). First, it must be acknowledged that interaction models demand a large number of data points and must be interpreted with caution. Furthermore, for the empirical analogue $L_{\max}$, the influence of PP was insignificant (Table 5). Nevertheless, the negative influence of high productivity in brackish waters on L_{∞} and ϕ' can be explained by its association with hypoxic episodes, where smaller body size provides a survival advantage due to a lower oxygen demand. The previously discussed Azov-Black Sea basin is a prime example of this phenomenon, as are other estuarine regions.

Basilone et al. (2004) evaluated the influence of productivity (measured by chlorophyll-a concentrations) on anchovy growth, reporting a negative relationship between maximum sizes and productivity, alongside a positive correlation between the growth rate constant k and productivity. However, in a later study, Basilone et al. (2017) reached contrasting conclusions, demonstrating higher early growth rates of anchovies in oligotrophic waters. By comparing the Tyrrhenian Sea and the Strait of Sicily, they emphasized the role of water transparency in enhancing anchovy foraging efficiency. Similarly, Bacha et al. (2010) observed a statistically significant negative correlation between length in the first year of life and chlorophyll concentration in the Mediterranean basin. Conversely, Basilone et al. (2020), while studying anchovy spawning ecology under upwelling conditions in the Strait of Sicily, found that increased productivity during upwelling events positively correlates with early growth rates.

Thus, the influence of productivity on growth and size remains a contentious issue. This influence is likely mediated by population density and size. In response to changes in productivity, rapid self-regulation of population size may occur to align with food availability (Murdoch, 1994), particularly in species with short life cycles, such as anchovies. As a result, the amount of food available per individual remains relatively stable, allowing life-history parameters (e.g. organism size and growth rate) to be maintained at consistent levels. This strategy enables the production of high-quality individuals even under low-productivity conditions, albeit with a smaller population size, and appears to be evolutionarily stable.

4.6 | Summary and implications for climate change

In summary, not all elements of the proposed hypothesis regarding the variability of growth and size of anchovies (within the framework of the TSR) are confirmed (Figure S1). Temperature appears to have the most significant influence, with maximum sizes, maturation sizes and the growth performance index showing the expected negative correlation with temperature. However, contrary to predictions, two

key patterns were observed: (1) a negative relationship was found between temperature and early growth rates, and (2) no significant association was detected between temperature and lifespan across the entire study range. This discrepancy suggests that early growth and lifespan may be influenced by temperature in different ways in some cases, or that other factors, such as food availability, predation pressure or fishing mortality, could obscure or override thermal effects. For example, in regions with high fishing pressure, the truncation of age structure may prevent the observation of temperature-related effects on lifespan (Anderson et al., 2008). Similarly, early growth may be more sensitive to local environmental conditions, such as the predator pressure or the availability of prey, rather than temperature alone (Peters, 1986). Additionally, a noticeable deviation from the general pattern of variability is observed within the Azov-Black Sea basin, suggesting a complex interplay of different environmental factors. Changes in size, growth and life strategies under the influence of temperature represent a trade-off between benefits and costs, suggesting that these changes cannot be universally explained by a single mechanism. It appears that variability within the framework of the TSR generally provides greater benefits; however, under certain conditions, such as the comparatively cold but unstable and hypoxic environment of the Azov Sea, the costs may outweigh the benefits.

Based on these findings, a decrease in anchovy size can be anticipated as climate warming progresses. However, given the existence of different anchovy ecotypes, characterized by varying resilience to changes in specific environmental parameters, their responses to changing environmental conditions may differ to some extent. Coastal anchovies, often found in more stable and warmer environments near river mouths (Catanese et al., 2017), may exhibit higher thermal tolerance but could be more sensitive to extreme temperature fluctuations. As global temperatures rise, this ecotype might experience reduced maximum sizes and earlier maturation, aligning with the observed negative correlation between temperature and size. However, their ability to adapt to warmer conditions could be limited if temperature changes exceed their thermal optima. Marine anchovies, inhabiting open waters with more variable temperatures, may demonstrate greater phenotypic plasticity in response to temperature changes. Their ability to thrive in oligotrophic and warmer waters suggests a broader thermal tolerance. However, prolonged exposure to higher temperatures could still lead to smaller body sizes and altered growth rates, as seen in the general trend of temperature-driven size reduction. Changes in salinity and productivity caused by warming may also exhibit ecotype-specific differences. Coastal anchovies are adapted to lower salinity levels near river mouths, where freshwater input dilutes seawater (Catanese et al., 2017). They may exhibit higher tolerance to salinity fluctuations but could be vulnerable to extreme changes, such as prolonged droughts or heavy rainfall, which alter salinity regimes. Marine anchovies, adapted to stable, higher-salinity environments, may be less tolerant of salinity fluctuations. Thus, a possible decrease in salinity may contribute more significantly to a reduction in their size compared to coastal form. Coastal anchovies thrive in high-productivity zones near river mouths, where nutrient input supports abundant prey. However, excessive productivity could

lead to hypoxic conditions, negatively affecting growth and survival. This ecotype may exhibit rapid population adjustments to align with food availability, maintaining stable growth. Marine anchovies, adapted to lower-productivity, oligotrophic waters, may rely on efficient foraging strategies to exploit sparse resources. They could benefit from moderate increases in productivity, such as during upwelling events, which may enhance early growth rates (Basilone et al., 2020). However, their growth and size may be less directly tied to productivity, as they are more influenced by temperature. It is important to remember that both ecotypes face trade-offs in responding to temperature, salinity, productivity and a multitude of other factors. Predicting exactly which life strategy will yield the greatest reproductive success and become fixed by selection in a particular population remains a challenging task at this time.

While the models used in this study provide valuable insights, several limitations must be acknowledged. First, they assume that SST, PP and salinity are the primary drivers of variation in life-history traits. However, unmeasured factors—including predation pressure, competition, fishing pressure and genetic differences between populations—could also influence these traits. Second, differences in sampling methods and measurement techniques may introduce bias. Third, historical data (particularly from older studies) may lack the precision of modern datasets, potentially affecting the reliability of the results. Fourth, the ability to capture the full complexity of life-history variability in the models, especially for interaction models with tensor terms, was limited by the restricted number of data points from different regions and environmental conditions.

To improve our understanding of how environmental factors influence anchovy life strategies, further monitoring of biological parameters across diverse environmental conditions is essential. This should be complemented by including additional environmental variables and more data points, especially from understudied regions such as the West African coast or the North Sea. Experimental studies are also needed to clarify the phenotypic effects associated with different environmental parameters. Furthermore, investigating the variability of life strategies and the mechanisms driving this variability requires a comprehensive approach, potentially incorporating mathematical frameworks such as the theory of games. Interdisciplinary methodologies could provide deeper insights into the evolutionary pressures and environmental interactions shaping anchovy populations, enhancing our ability to predict outcomes in the context of ongoing climate change and ecological shifts. Ultimately, research integrating these factors and approaches will be crucial for developing effective management practices for anchovy fisheries.

ACKNOWLEDGEMENTS

The author sincerely thanks the handling editor, Dr. Sofia Graça Aranha, for her thoughtful guidance and constructive recommendations, which greatly improved the clarity and analytical strength of this manuscript. The author is also grateful to the anonymous reviewers for their careful evaluation and insightful comments that contributed to refining the methodology and interpretation of the results. Their efforts and expertise are deeply appreciated.

FUNDING INFORMATION

This work was carried out within the framework of IBSS state research assignment “Biodiversity as the basis for the sustainable functioning of marine ecosystems, criteria and scientific principles for its conservation” (No. 124022400148-4).

CONFLICT OF INTEREST STATEMENT

The author declares that he has no financial or non-financial interests that are directly or indirectly related to the work submitted for publication.

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REFERENCES

- Altukhov, Y. P. (2001). Genetic consequences of selective fishing and fish farming. *Voprosy Rybolovstva*, 2(4), 62–603.
- Amponsah, S. K., Ofori-Danson, P. K., & Nunoo, F. K. (2016). Population dynamics of *Engraulis encrasicolus* (Linnaeus, 1758) within Ghana's coastal waters. *International Journal of Fisheries and Aquatic Studies*, 4(5), 258–263. <https://csirspace.foodresearchgh.site/handle/123456789/469>
- Anderson, C. N., Hsieh, C. H., Sandin, S. A., Hewitt, R., Hollowed, A., Beddington, J., & Sugihara, G. (2008). Why fishing magnifies fluctuations in fish abundance. *Nature*, 452(7189), 835–839. <https://doi.org/10.1038/nature06851>
- AquaMaps. (2019). Computer generated distribution maps for *Engraulis encrasicolus* (European anchovy), with modelled year 2050 native range map based on IPCC RCP8.5 emissions scenario. Available at <https://www.aquamaps.org> (last accessed 19 September 2024)
- Atkinson, D. (1994). Temperature and organism size – a biological law for ectotherms? *Advances in Ecological Research*, 25, 1–58.
- Audzijonyte, A., Richards, S. A., Stuart-Smith, R. D., Pecl, G., Edgar, G. J., Barrett, N. S., Payne, N., & Blanchard, J. L. (2020). Fish body sizes change with temperature but not all species shrink with warming. *Nature Ecology & Evolution*, 4, 809–814. <https://doi.org/10.1038/s41559-020-1171-0>
- Bacha, M., Moali, A., Benmansour, N. E., Brylinski, J. M., Mahe, K., & Amara, R. (2010). Relationships between age, growth, diet and environmental parameters for anchovy (*Engraulis encrasicolus* L.) in the Bay of Bénisaf (SW Mediterranean, west Algerian coast). *Cybio*, 34(1), 47–57. <https://archimer.ifremer.fr/doc/00011/12212/>
- Bartoń, K. (2025). MuMIn: Multi-Model Inference. <https://doi.org/10.32614/CRAN.package.MuMIn>
- Basilone, G., Ferreri, R., Barra, M., Bonanno, A., Pulizzi, M., Gargano, A., & Aronica, S. (2020). Spawning ecology of the European anchovy (*Engraulis encrasicolus*) in the Strait of Sicily: Linking variations of zooplankton prey, fish density, growth, and reproduction in an upwelling system. *Progress in Oceanography*, 184, 102330. <https://doi.org/10.1016/j.poccean.2020.102330>
- Basilone, G., Guisande, C., Patti, B., Mazzola, S., Cuttitta, A., Bonanno, A., & Kallianiotis, A. (2004). Linking habitat conditions and growth in the European anchovy (*Engraulis encrasicolus*). *Fisheries Research*, 68(1–3), 9–19. <https://doi.org/10.1016/j.fishres.2004.02.012>
- Basilone, G., Mangano, S., Pulizzi, M., Fontana, I., Giacalone, G., Ferreri, R., Gargano, A., Aronica, S., Barra, M., Genovese, S., et al. (2017). European anchovy (*Engraulis encrasicolus*) age structure and growth rate in two contrasted areas of the Mediterranean Sea: the paradox of faster growth in oligotrophic seas. *Mediterranean Marine Science*, 18(3), 504–516. <https://doi.org/10.12681/mms.2059>
- Bellido, J. M., Pierce, G. J., Romero, J. L., & Millan, M. (2000). Use of frequency analysis methods to estimate growth of anchovy (*Engraulis encrasicolus* L. 1758) in the Gulf of Cadiz (SW Spain). *Fisheries Research*, 48(2), 107–115. [https://doi.org/10.1016/S0165-7836\(00\)00183-1](https://doi.org/10.1016/S0165-7836(00)00183-1)
- Bembo, D. G., Carvalho, G. R., Cingolani, N., & Pitcher, T. J. (1996). Stock discrimination among European anchovies, *Engraulis encrasicolus*, by means of PCR-amplified mitochondrial DNA analysis. *Fishery Bulletin*, 94, 31–40.
- Berdnikov, S. V., Dashkevich, L. V., & Kulygin, V. V. (2019). Climatic conditions and hydrological regime of the sea of Azov in the XX – early XXI centuries. *Водные биоресурсы и среда обитания*, 2(2), 7–19. https://doi.org/10.47921/2619-1024_2019_2_2_7
- Berdnikov, S. V., Kulygin, V. V., & Dashkevich, L. V. (2019). Seasonal and long-term dynamics of primary production in the Sea of Azov from MERIS data. *Sovremennye Problemy Distsionnogo Zondirovaniia Zemli iz Kosmosa*, 16(2), 173–185. <https://doi.org/10.21046/2070-7401-2019-16-2-173-185>
- Blaszczak, P. (1999). Length and age composition and growth of anchovy *Engraulis encrasicolus* (L. 1758) caught in the East Schelde, The Netherlands, in 1987. *Folia Universitatis Agriculturae Stetinensis*, 192, 5–11.
- Bœuf, G., & Payan, P. (2001). How should salinity influence fish growth? *Comparative Biochemistry and Physiology Part C: Toxicology & Pharmacology*, 130(4), 411–423. [https://doi.org/10.1016/S1532-0456\(01\)00268-X](https://doi.org/10.1016/S1532-0456(01)00268-X)
- Bosc, E., Bricaud, A., & Antoine, D. (2004). Seasonal and interannual variability in algal biomass and primary production in the Mediterranean Sea, as derived from 4 years of SeaWiFS observations. *Global Biogeochemical Cycles*, 18, GB1005. <https://doi.org/10.1029/2003GB002034>
- Breheny, P., & Burchett, W. (2017). Visualization of Regression Models Using visreg. *The R Journal*, 9(2), 56–71. <https://doi.org/10.32614/RJ-2017-046>
- Brett, J. R. (1979). Environmental factors and growth. In W. S. Hoar, D. J. Randall & J. R. Brett (Eds.), *Fish Physiology: Bioenergetics and Growth* (Vol. 8, pp. 599–675). Academic Press.
- Catanese, G., Watteaux, R., Montes, I., Barra, M., Rumolo, P., Borme, D., & Procaccini, G. (2017). Insights on the drivers of genetic divergence in the European anchovy. *Scientific Reports*, 7(1), 4180. <https://doi.org/10.1038/s41598-017-03926-z>
- Cendrero, O., Cort, J. L., & de Camas, E. (1981). Revisión de algunos datos sobre la biología de la anchoa, *Engraulis encrasicolus* (L.) del Mar Cantábrico. *Boletín del Instituto Español de Oceanografía*, 6, 117–124.
- Chashchin, A. K. (1996). The Black Sea populations of anchovy. *Scientia Marina*, 60, 219–225.
- Chesalin, M., Nikolsky, V., & Yuneva, T. (2020). Biological Characteristics of Azov Anchovy (*Engraulis encrasicolus maeoticus* A.) in 2016–2017 and 2017–2018 Fishing Seasons. *Turkish Journal of Fisheries and Aquatic Sciences*, 20, 559–570. https://doi.org/10.4194/1303-2712-v20_7_06
- Cheung, W. W., Sarmiento, J. L., Dunne, J., Frölicher, T. L., Lam, V. W., Deng Palomares, M. L., & Pauly, D. (2013). Shrinking of fishes exacerbates impacts of global ocean changes on marine ecosystems. *Nature Climate Change*, 3(3), 254–258. <https://doi.org/10.1038/nclimate1691>
- Colloca, F., Cardinale, M., Maynou, F., Giannoulaki, M., Scarcella, G., Jenko, K., Bellido, J. M., & Fiorentino, F. (2013). Rebuilding Mediterranean fisheries: a new paradigm for ecological sustainability. *Fish and fisheries*, 14(1), 89–109. <https://doi.org/10.1111/j.1467-2979.2011.00453.x>
- Dawkins, R. (2016). *The selfish gene*. Oxford University Press.
- El-Betar, T., Shalloof, K., El-Far, A., Fetouh, M., Aly, W., & El-Ganiny, A. (2023). Population structure of the European anchovy, *Engraulis encrasicolus* (Linnaeus, 1758) in Lake Manzala, Egypt. *International Journal of Aquatic Biology*, 11(6), 505–512. <https://doi.org/10.22034/ijab.v11i6.1889>

- FAO. (2018). *Report of the FAO Working Group on the Assessment of Small Pelagic Fish off Northwest Africa. Nouadhibou, Mauritania, 22–27 May 2017*. FAO Fisheries and Aquaculture Report No. 1221. FAO.
- FAO. (2023). *The State of Mediterranean and Black Sea Fisheries 2023*. FAO. <https://doi.org/10.4060/cc8888en>
- Ferreri, R., Genovese, S., Barra, M., Biagiotti, I., Bourdeix, J. H., De Felice, A., Gašparević, D. E. N. I. S., Hattab, T. A. R. E. K., Iglesias, M. A. G. D. A. L. E. N. A., Juretić, T. E. A., Leonori, I. O. L. E., Malavolti, S. A. R. A., Raykov, V. I. O. L. I. N. S. T., Sarau, C. L. A. I. R. E., Tičina, V. J. E. K. O. S. L. A. V., Ventero, A. N. A., & Basilone, G. (2021). Variability in size at maturity of the European anchovy (*Engraulis encrasicolus*) in the Mediterranean Sea. *Mediterranean Marine Science*, 22(4), 858–870. <https://doi.org/10.12681/mms.25995>
- Finenko, Z. Z., Suslin, V. V., & Churilova, T. Y. (2010). Estimation of phytoplankton productivity in the Black Sea based on satellite data. *Doklady Biological Sciences*, 432(1), 233–236. <https://doi.org/10.1134/S0012496610030191>
- Fischer, G., Ratmeyer, V., & Wefer, G. (2000). Organic carbon fluxes in the Atlantic and the Southern Ocean: relationship to primary production compiled from satellite radiometer data. *Deep Sea Research Part II: Topical Studies in Oceanography*, 47(9–11), 1961–1997. [https://doi.org/10.1016/S0967-0645\(00\)00013-8](https://doi.org/10.1016/S0967-0645(00)00013-8)
- Froese, R., & Pauly, D. (2024). FishBase. Available at: <http://www.fishbase.org> (last accessed 2 April 2024)
- Gardner, J. L., Peters, A., Kearney, M. R., Joseph, L., & Heinsohn, R. (2011). Declining body size: a third universal response to warming? *Trends in ecology and evolution*, 26(6), 285–291. <https://doi.org/10.1016/j.tree.2011.03.005>
- Gücü, A. C., Genç, Y., Dağtekin, M., Sakınan, S., Ak, O., Ok, M., & Aydın, İ. (2017). On Black Sea anchovy and its fishery. *Reviews in Fisheries Science & Aquaculture*, 25(3), 230–244. <https://doi.org/10.1080/23308249.2016.1276152>
- Guérault, D., & Avrilla, J. L. (1974). L'anchois du golfe de Gascogne: taille, âge, croissance. *Journal du Conseil. Conseil Permanent International pour l'Exploration de la Mer*, CM 1974/J:17.
- Hattab, T., Gucu, A., Ventero, A., De Felice, A., Machias, A., Sarau, C., Gucu, A. L. I., Ventero, A., Gašparević, D., Basilone, G., Costantini, I., Leonori, I., Bourdeix, J.-h., Iglesias, M., Barra, M., Giannoulaki, M., Ferreri, R., El Ayoubi, S., Malavolti, S., ... Certain, G. (2021). Temperature strongly correlates with regional patterns of body size variation in Mediterranean small pelagic fish species. *Mediterranean Marine Science*, 22(4), 800. <https://doi.org/10.12681/mms.26525>
- Hémida, F. (1987). Contribution à l'étude de l'anchois: *Engraulis encrasicolus* (Linné, 1758) dans la région d'Alger. Biologie et exploitation, Thèse de Magister, Alger.
- Holt, J., Butenschön, M., Wakelin, S. L., Artioli, Y., & Allen, J. I. (2012). Oceanic controls on the primary production of the northwest European continental shelf: model experiments under recent past conditions and a potential future scenario. *Biogeosciences*, 9(1), 97–117. <https://doi.org/10.5194/bg-9-97-2012>
- ICES. (2010). Report of the Workshop on Age reading of European anchovy (WKARA), 9–13 November 2009, Sicily, Italy. ICES CM 2009/ACOM.43.
- Karacam, H., & Düzgünes, E. (1990). Age, growth and meat yield of the European anchovy (*Engraulis encrasicolus*, L. 1758) in the Black Sea. *Fisheries Research*, 9(2), 181–186.
- Khemiri, S., Gaamour, A., Meunier, F. J., & Zylberberg, L. (2007). Age and growth of *Engraulis encrasicolus* (Clupeiforme: Engraulidae) in the Tunisian waters. *Cahiers de Biologie Marine*, 48(3), 259–269.
- Kozłowski, J., Czarnotański, M., & Dańko, M. (2004). Can optimal resource allocation models explain why ectotherms grow larger in cold? *Integrative and Comparative Biology*, 44(6), 480–493. <https://doi.org/10.1093/icb/44.6.480>
- Machias, A., Somarakis, S., Kapadagakis, P., Drakopoulos, P., Giannoulaki, M., Maraveya, E., Manousakis, L., Vatsos, D., Tsimenidou, C., & Tsimenides, N. (2000). Evaluation of the southern Greek anchovy stocks. *Final report DGXIV contract no. 97/0048*.
- Mayorova, A. A. (1951). *Biology and fishery of the Black Sea anchovy*. Crymizdat.
- Morales-Nin, B., & Perterra, J. P. (1990). Growth rates of the anchovy *Engraulis encrasicolus* and the sardine *Sardina pilchardus* in the north-western Mediterranean Sea. *Marine Biology*, 107, 349–356. <https://doi.org/10.1007/BF01319836>
- Murdoch, W. W. (1994). Population regulation in theory and practice. *Ecology*, 75(2), 271–287. <https://doi.org/10.2307/1939533>
- Palomera, I., Olivar, M. P., Salat, J., Sabatés, A., Coll, M., García, A., & Morales-Nin, B. (2007). Small pelagic fish in the NW Mediterranean Sea: An ecological review. *Progress in Oceanography*, 74, 377–396. <https://doi.org/10.1016/j.pcean.2007.04.012>
- Pauly, D. (1978). A preliminary compilation of fish length growth parameters. *Berichte des Instituts für Meereskunde an der Christian-Albrechts-Universität Kiel*, 55, 1–200.
- Pauly, D. (1998). Tropical fishes: Patterns and propensities. *Journal of Fish Biology*, 53, 1–17. <https://doi.org/10.1006/jfbi.1998.0810>
- Pauly, D. (2021). The gill-oxygen limitation theory (GOLT) and its critics. *Science Advances*, 7(2), eabc6050. <https://doi.org/10.1126/sciadv.abc6050>
- Pauly, D., & Cheung, W. W. (2018). Sound physiological knowledge and principles in modeling shrinking of fishes under climate change. *Global Change Biology*, 24(1), e15–e26. <https://doi.org/10.1111/gcb.13831>
- Pauly, D., Moreau, J., & Prein, M. (1988). A comparison of overall growth performance of tilapia in open waters and aquaculture. In *ICLARM Conf. Proc. (Vol. 15, pp. 469–479)*. International Center for Living Aquatic Resources Management.
- Peters, R. H. (1986). *The ecological implications of body size*. Cambridge University Press.
- Quinn, G. P., & Keough, M. J. (2002). *Experimental design and data analysis for biologists*. Cambridge University Press.
- R Core Team. (2025). R: A language and environment for statistical computing. R Foundation for Statistical Computing. <https://www.R-project.org/>
- Ramos, S., & Santos, P. (1999). Crescimento do biqueirao (*Engraulis encrasicolus* Linnaeus, 1758) na costa norte de Portugal. *Revista de Biologia*, 17, 211–216.
- Ricker, W. E. (2010). Computation and interpretation of biological statistics of fish populations. In *Bulletin of the Fisheries Research Board of Canada (Vol. 191)*. Blackburn Press.
- Sakalli, A., & Baştuta, N. (2018). Sea surface temperature change in the Black Sea under climate change: A simulation of the sea surface temperature up to 2100. *International Journal of Climatology*, 38(13), 4687–4698. <https://doi.org/10.1002/joc.5688>
- Samsun, O., Samsun, N., Kalayci, F., & Bilgin, S. (2006). A Study on Recent Variations in the Population Structure of European Anchovy (*Engraulis encrasicolus* L., 1758) in the Southern Black Sea. *Ege Journal of Fisheries and Aquatic Sciences*, 23(3), 301–306. <http://www.egejfas.org/en/download/article-file/57631>
- Samsun, O., Samsun, N., & Karamollaoğlu, A. C. (2004). Age, growth, and mortality rates of the European anchovy (*Engraulis encrasicolus* L. 1758) off the Turkish Black Sea coast. *Turkish Journal of Veterinary & Animal Sciences*, 28(5), 901–910. <https://journals.tubitak.gov.tr/veterinary/vol28/iss5/18>
- Schultz, E. T., & Conover, D. O. (1997). Latitudinal differences in somatic energy storage: adaptive responses to seasonality in an estuarine fish (Atherinidae: *Menidia menidia*). *Oecologia*, 109(4), 516–529.
- Sheridan, J. A., & Bickford, D. (2011). Shrinking body size as an ecological response to climate change. *Nature climate change*, 1(8), 401–406. <https://doi.org/10.1038/nclimate1259>
- Sinović, G. (2000). Anchovy, *Engraulis encrasicolus* (Linnaeus, 1758): biology, population dynamics and fisheries case study. *Acta Adriatica*, 41(1), 3–53.

- Sinovčić, G. (2004). Growth and length–weight relationship of the juvenile anchovy, *Engraulis encrasicolus*, in the nursery ground (Zrmanja River estuary–eastern Adriatic Sea). *Journal of applied ichthyology*, 20(1), 79–80. <https://doi.org/10.1046/j.0175-8659.2003.00517.x>
- STECF. (2009a). In M. Cardinale, H. J. Raetz, & A. Cheilari (Eds.), *Report of the SGMED-09-02 Working Group on the Mediterranean Part I*. Office for Official Publications of the European Communities.
- STECF. (2009b). In M. Cardinale, H. J. Raetz, & A. Cheilari (Eds.), *Report of the SGMED-09-03, Working Group on the Mediterranean - Part II*. Office for Official Publications of the European Communities.
- Taboada, F. G., Chust, G., Santos Mocoroa, M., Aldanondo, N., Fontán, A., Cotano, U., Álvarez, P., Erauskin-Extramiana, M., Irigoien, X., Fernandes-Salvador, J. A., Boyra, G., Uriarte, A., & Ibaibarriaga, L. (2024). Shrinking body size of European anchovy in the Bay of Biscay. *Global Change Biology*, 30(1), e17047. <https://doi.org/10.1111/gcb.17047>
- Uriarte, A., Rico, I., Villamor, B., Duhamel, E., Dueñas, C., Aldanondo, N., & Cotano, U. (2016). Validation of age determination using otoliths of the European anchovy (*Engraulis encrasicolus* L.) in the Bay of Biscay. *Marine and Freshwater Research*, 67(7), 951–966. <https://doi.org/10.1071/MF15092>
- Verberk, W. C., Atkinson, D., Hoefnagel, K. N., Hirst, A. G., Horne, C. R., & Siepel, H. (2021). Shrinking body sizes in response to warming: explanations for the temperature–size rule with special emphasis on the role of oxygen. *Biological Reviews*, 96(1), 247–268. <https://doi.org/10.1111/brv.12653>
- Vodiasova, E. A., & Abramson, N. I. (2017). Genetic variability of anchovy in the Azov-Black Sea basin. *Russian Journal of Genetics*, 53, 680–687. <https://doi.org/10.1134/S1022795417060138>
- Volovik, S. P., & Kozlitina, S. V. (1983). Assessment of the potential catch of the azov anchovy, *Engraulis encrasicolus* (Engraulidae), in relation to growth and structure of its population. *Journal of Ichthyology*, 23(1), 26–38.
- Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L. D., François, R., Grolemund, G., Hayes, A., McGowan, L., Henry, L., Hester, J., Kuhn, M., Pedersen, T., Miller, E., Bache, S., Müller, K., Ooms, J., Robinson, D., Seidel, D., ... Yutani, H. (2019). Welcome to the tidyverse. *Journal of Open Source Software*, 4(43), 1686. <https://doi.org/10.21105/joss.01686>
- Wood, S. N. (2011). Fast stable restricted maximum likelihood and marginal likelihood estimation of semiparametric generalized linear models. *Journal of the Royal Statistical Society. Series B, Statistical Methodology*, 73(1), 3–36.
- World Ocean Atlas. (2025). Annual salinity statistical mean at the surface: World Wide Web Electronic Publication. Available at: <https://www.ncei.noaa.gov/access/world-ocean-atlas-2023f/bin/woa23f.pl> (last accessed 13th May 2025)
- Yankova, M. (2014). Preliminary estimates of the population parameters of four species in the Bulgarian Black Sea coast. *International Journal of Latest Research in Science and Technology*, 3(5), 46–52.
- Zuur, A. F., Ieno, E. N., Walker, N. J., Saveliev, A. A., & Smith, G. M. (2009). *Mixed effects models and extensions in ecology with R* (Vol. 574, p. 574). Springer.
- Zuyev, G., & Skuratovskaya, E. (2023). Population Structure of European Anchovy *Engraulis encrasicolus* (L.) (Engraulidae: Pisces) in the Azov-Black Sea Basin. *Thalassas*, 39(1), 115–124. <https://doi.org/10.1007/s41208-023-00529-6>
- Zuyev, G. V. (2019). Current population structure of European anchovy *Engraulis encrasicolus* L. (Engraulidae: Pisces) in the Sea of Azov–Black Sea basin and history of its formation. *Marine Biological Journal*, 4(1), 45–62. <https://doi.org/10.21072/mbj.2019.04.1.05>

SUPPORTING INFORMATION

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How to cite this article: Kutsyn, D. (2025). Ecological patterns of variability in the life history of the European anchovy (*Engraulis encrasicolus*). *Journal of Fish Biology*, 1–16. <https://doi.org/10.1111/jfb.70314>